

PRINT Name: _____

Test Version: 000 One-Answer Multiple Choice 25 Questions – 7.5 of 15%

☞ Fill in the bubbles for your Name and Answers with pencil only, no pen.

☞ Manage your time. Answer questions you know, first. One Answer per question.

1. What is Reinforcement Learning (RL)?
 - a. RL is a third type of machine learning, along with supervised learning and unsupervised learning.
 - b. None of these answers.
 - c. All of these answers.
 - d. RL is unsupervised machine learning used for control applications.
 - e. RL is supervised machine learning for systems that play games.
2. Which of the following statements is true in the context of Reinforcement Learning?
 - a. None of these answers.
 - b. Q-learning is a process for learning the action-value function.
 - c. All of these answers.
 - d. Q-learning cannot be used directly on an environment with continuous actions or continuous states.
 - e. Q-learning can be applied to environments with manageable numbers of discrete actions and discrete states.
3. What is a Markov state?
 - a. Markov states are states that form a single deterministic chain without cycles.
 - b. None of these answers.
 - c. Intuitively, a Markov state has the property that all its previous states determine whether it is a goal state.
 - d. All of these answers.
 - e. Intuitively, a Markov state has the property that its subsequent states do not depend on its previous states.
4. Which of the following methods of a Gymnasium environment would be called when an episode ends?

a. All of these answers.	b. None of these answers.
c. reset	d. render
e. step	
5. What is a typical way that Stable-Baselines3 is used?
 - a. All of these answers.
 - b. Train a model, then implement an environment based on that model
 - c. Train an environment, then create a model to be used instead of the environment
 - d. Create a model based on an environment, then train the model
 - e. None of these answers.

6. What is the Reward Hypothesis of Reinforcement Learning?
 - a. None of these answers.
 - b. The Reward Hypothesis basically states that some goals cannot be thought of as maximizing the expected cumulative value of a scalar reward function.
 - c. The Reward Hypothesis basically states that all goals can be thought of as maximizing the expected cumulative value of a scalar reward function.
 - d. The Reward Hypothesis basically states that all goals can be thought of as minimizing the the number of steps to maximize a scalar reward function.
 - e. The Reward Hypothesis basically states that some goals cannot be thought of as minimizing the the number of steps to maximize a scalar reward function.
7. Which of the following is a part of every Gymnasium environment?

a. A step method	b. A neural network
c. None of these answers	d. All of these answers
e. A policy for taking action	
8. What is the relationship between Reinforcement Learning (RL) and Markov Decision Processes (MDPs)?
 - a. MDPs are known specific strategies developed for playing games like chess, go, and video games played by RL systems.
 - b. MDPs are a mathematical model of the sequential decision making processes addressed by RL.
 - c. None of these answers.
 - d. MDPs are a component of the software used to implement video games.
 - e. All of these answers.
9. Which of the following statements is true about the Bellman equation in Reinforcement Learning?
 - a. None of these answers.
 - b. It breaks the problem of determining the value of a state into smaller problems recursively.
 - c. All of these answers.
 - d. It forms the mathematical basis for the Q-Learning algorithm in Reinforcement Learning.
 - e. It expresses the relationship between the value of a state or a state-action pair, and the value of the successor states.
10. Which of the following is a feature of Stable-Baselines3?
 - a. All of these answers.
 - b. Environment classes for standard Reinforcement Learning problems
 - c. None of these answers.
 - d. Callback methods for implementing monitoring, progress bars, and more.
 - e. Rendering methods based on PyGame

11. Which of the following statements is true in the context of Reinforcement Learning?
 - a. The results of an action are determined by the agent rather than the environment.
 - b. The value function and policy function are implemented in the agent rather than the environment.
 - c. The policy function is implemented in the environment rather than the agent.
 - d. The value function is implemented in the environment rather than the agent.
 - e. None of these answers.
12. What is Stable-Baselines3?
 - a. None of these answers.
 - b. A set of video game environment implementations.
 - c. A set of standard Reinforcement Learning problems.
 - d. All of these answers.
 - e. A set of algorithm implementations for Reinforcement Learning.
13. Which of the following is implemented in Stable-Baselines3?
 - a. None of these answers.
 - b. GridWorld environment
 - c. DQN algorithm
 - d. All of these answers.
 - e. CliffWalking environment
14. Which of the following is a step in creating and using a custom Gymnasium environment?
 - a. None of these answers.
 - b. Instantiate the environment with `gymnasium.make`.
 - c. Register the name of the custom environment in the registry.
 - d. Define a class that inherits from the Gymnasium Env class.
 - e. All of these answers.
15. What is the difference between an action value function and a state value function?
 - a. State value functions return total reward to termination, and action-value functions return immediate reward of taking the action.
 - b. Action value functions return the average reward for taking an action, and State value functions return a state's average total future reward.
 - c. None of these answers.
 - d. State value functions take a state, and action value functions take just actions.
 - e. Action value functions take state-action pairs, whereas state value functions take just states.
16. What is meant by "episode" in Reinforcement Learning?
 - a. None of these answers.
 - b. An episode is a single run that cycles through the terminal state multiple times.
 - c. An episode is a single run from the starting state to a terminal (or truncated) state.
 - d. An episode is a single cycle of performing an action, receiving a reward, and observing the resulting state.
 - e. An episode is a number equal to the number of steps actually taken to reach the terminal state.

17. What are vectorized environments in the context of Stable-Baselines3?
 - a. They are used for representing multi-dimensional observations.
 - b. They are a method for training an agent on more than one copy of an environment.
 - c. All of these answers.
 - d. They are a method for training multiple agents on a single environment.
 - e. None of these answers.
18. What is a Policy in Reinforcement Learning?
 - a. The Policy is a function that assigns a value to each action-state pair.
 - b. None of these answers.
 - c. All of these answers.
 - d. The Policy is a table that assigns a value to each action.
 - e. The Policy is a function that determines the probability of an agent taking an action.
19. What is a condition for applying Q-learning to a Reinforcement Learning problem?
 - a. The optimal value function must be known.
 - b. The complete set of actions must be known.
 - c. The complete set of possible states must be known.
 - d. The complete set of actions and the complete set of possible states must be small enough to implement a q-table.
 - e. None of these answers.
20. Why role does the discount factor γ play in Reinforcement Learning?
 - a. γ addresses the problem of infinite cumulative rewards in non-terminating processes.
 - b. None of these answers.
 - c. γ represents the total discount which is subtracted from the total reward after an episode.
 - d. γ determines how many times an action is chosen randomly during training.
 - e. γ represents the weighting of the current goal of a Reinforcement Learning problem.
21. What does "greedy" mean in the context of Reinforcement Learning?
 - a. None of these answers.
 - b. It implies a policy where future reward is considered over immediate reward.
 - c. It implies a policy that tries to maximize future reward.
 - d. It implies a policy where immediate reward is considered over future reward.
 - e. It implies a policy that tries to maximize total reward.

22. What is Gymnasium in the context of Reinforcement Learning?
- a. None of these answers.
 - b. A set of standard algorithms for solving Reinforcement Learning problems.
 - c. An algorithm for determining optimal hyper-parameter values.
 - d. A library and API standard for environments that includes a number of pre-defined reference environments.
 - e. All of these answers.
23. What is a Value Function in Reinforcement Learning?
- a. A Value Function gives a measure of the expected total reward given a state or state-action pair
 - b. A Value Function gives a measure of the expected total number of steps to maximize reward.
 - c. None of these answers.
 - d. All of these answers.
 - e. A Value Function gives a measure of the expected total reward of an episode.
24. Which of the following can be considered primary aspects of a Reinforcement Learning problem setup?
- a. Values, Step function, and Actions.
 - b. None of these answers.
 - c. Reward, Environment, and Models.
 - d. Agent, Values, and Step function.
 - e. Agent, Environment, and Reward.
25. Which of the following methods of a Gymnasium environment returns an observation?
- a. make
 - b. step
 - c. None of these answers.
 - d. render
 - e. All of these answers.

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